

THE STANDARD ERROR OF ESTIMATED INTENSITY FUNCTIONS OF STATIONARY POINT PROCESSES

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1. Introduction

There are two estimators available for the intensity function of a stationary point process from observed lengths of intervals between points (Cox, 1965; Linhart, 1964). Both methods can obviously also be used to estimate the closely related covariance density function, or the statistical anomaly function. We give the approximate variance for both estimators and consider its estimation. The variance is approximate because both estimators are functions of observed lengths (for terminology see Linhart, 1970). The process of lengths is not stationary, but in our derivation stationarity is assumed. The error incurred is small since statistics calculated from observed lengths behave almost as if they were calculated from distances, and the process of distances is stationary. For a more thorough discussion of this see Linhart and Zucchini (1967) and Linhart (1970, 1972).

The results obtained here follow immediately from those reported in Linhart, (1975). The notation is explained in Section 2 of that paper.

2. The variance of estimators of the intensity function

If $X_1, X_2, \dots, X_n$ is a given realisation of size n from the process of lengths, the two estimators of the intensity function are

$$(1) \quad \hat{h}_1(u + \Delta u/2) = \sum_{i=1}^n n f_i(u) \quad (\text{Linhart, 1964})$$

and

$$(2) \quad \hat{h}_2(u + \Delta u/2) = (t_0/n (t_0 - u - \Delta u/2)) \sum_{i=1}^n (n-i+1) n f_i(u),$$

where $N \leq n$, and $t_0 = \sum_{i=1}^n X_i$ (Cox, 1965).

We need therefore the variances of $\sum_{i=1}^N n f_i(u)$ and of $\sum_{i=1}^n (n-i+1) n f_i(u)$, which are easily obtained from the covariance of $n f_r(u)$ and $n f_s(u)$, which is given in Linhart (1975):

$$\begin{aligned}
 (3) \quad \text{Var} \left(\sum_{i=1}^N n f_i(u) \right) &= 2 \sum_{s < r}^N (1/(n-r+1)(n-s+1)) \sum_{\nu=-(n-r)}^{n-s} n^*(n,r,s,\nu) f_{r(\nu)s}(u,u) \\
 &+ \sum_{s=1}^N (1/(n-s+1)^2) \sum_{\nu=-(n-s)}^{n-s} n^*(n,s,s,\nu) f_{s(\nu)s}(u,u) \\
 &- \sum_{r,s=1}^N f_r(u) f_s(u).
 \end{aligned}$$

A similar expression holds for

$$\text{Var} \left(\sum_{i=1}^n (n-i+1) n f_i(u) \right).$$

3. The standard error of estimators of the intensity function

In estimating the variances obtained in Section 2, the difficulties discussed in Linhart (1975) arise again: in general there are intolerably many parameters. One possibility to reduce the number of parameters is again to assume that X_i and X_j are independent for $|i-j| > d$, $d=0, 1, 2, \dots$

Under this assumption one gets

$$\begin{aligned}
 (4) \quad \text{Var} \left(\sum_{i=1}^N n f_i(u) \right) &= \sum_{r,s=1}^N \left(\frac{1}{(n-r+1)(n-s+1)} \sum_{\nu=-(d+s-1)}^{d+r-1} n^*(n,r,s,\nu) f_{r(\nu)s}(u,u) \right. \\
 &\left. + K(n,r,s,d) f_r(u) f_s(u) \right),
 \end{aligned}$$

$$\text{and a similar result for } \text{Var} \left(\sum_{i=1}^n (n-i+1) n f_i(u) \right).$$

The estimators are obtained by replacing $f_{r(\nu)s}(u,u)$ by $n f_{r(\nu)s}(u,u)$ and $f_r(u)$ by $n f_r(u)$. One gets

$$\begin{aligned}
 (5) \quad \hat{\text{Var}}(\hat{h}_1(u + \Delta u/2)) &= \frac{1}{(\Delta u)^2} \sum_{r,s=1}^{N'} \frac{1}{(n-r+1)(n-s+1)} \left(\sum_{\nu=-(d+s-1)}^{d+r-1} z(r,s;\nu;u,u) \right. \\
 &\left. + K(n,r,s,d) w(r;u) w(s;u) \right), \quad \text{and}
 \end{aligned}$$

$$\begin{aligned}
 (6) \quad \hat{\text{Var}}(\hat{h}_2(u + \Delta u/2)) &= (t_0/n(t_0 - u - \Delta u/2))^2 \frac{1}{(\Delta u)^2} \sum_{r,s=1}^{N'} \left(\sum_{\nu=-(d+s-1)}^{d+r-1} z(r,s;\nu;u,u) \right. \\
 &\left. + K(n,r,s,d) w(r;u) w(s;u) \right).
 \end{aligned}$$

In the case of $\hat{h}_1$ it is advisable to use $N' < N$ to reduce the number of parameters estimated, in the case of $\hat{h}_2$ n must in any case be replaced by $N' < n$ if one wants fewer parameters than observations.

4. Experiments

To gain insight into the performance of the estimators (5) and (6), Monte Carlo experiments were carried out. The point processes used were Poisson, uniform renewal and various normal autoregressive moving average processes (Linhart and Zucchini, 1972).

Of particular interest was the influence of d on the mean square error of the estimators. It was found that $d=0$ and $d=1$ are usually optimal choices, unless the lag one correlation or the sample size, or both, are very large. Values of d which are too large can increase the mean squared error disastrously especially if the sample size is small.

The table gives a few typical results. They were computed using samples of size n from a normal autoregressive process of order one with parameter ϕ , variance 1,04 (for $\phi = .2$) and 2,78 (for $\phi = .8$) and a mean which is three times the standard deviation (i.e. 3,06 and 5,01 respectively). The empirical variance was calculated from 200 independent estimators (2) of the intensity function at argument 3,06 and 5,01 respectively; Δu was $1/2$. Twenty independent estimates (6) were obtained, and the mean and variance of the distribution of this estimator were estimated in order to estimate the bias (empirical variance minus mean) and the mean square error ((bias)² plus variance).

The table gives also the relative error in per cent: $100 (\text{mean square error})^{1/2} / (\text{empirical variance})$, all calculated for $d=1$.

Zucchini (1973) derived an estimator of the variance of Cox's estimator (2) under the hypothesis that the point process is Poisson:

$$\hat{\text{Var}}_p(\hat{h}_2(u + \Delta u/2)) = n(4t_0 - 4u - 3\Delta u/4)/t_0(t_0 - u - \Delta u/2)^2 + 1/\Delta u(t_0 - u - \Delta u/2)$$

This is a convenient rough approximation for the standard error of $\hat{h}_2$. The table lists these approximations together with the relative errors, which in this case were taken as $100 (\text{empirical variance minus } \hat{\text{Var}}_p) / (\text{empirical variance})$.

As can be seen the relative errors of estimators (6) are tolerable for the sample sizes used here. The given errors all belong to $d=1$, there would be little change for $d=0$ or $d=2$, but large values of d can be damaging. Taking, for example, $d=5$ for $\phi = .2$, $n=100$, results in a relative error of 258 per cent. Zucchini's estimators under the Poisson hypothesis are certainly very good approximations, here they are even better for $\phi = .8$ than the estimator (6), but their performance depends obviously on the type of point process on hand. (For normal moving average processes the relative errors of $\hat{\text{Var}}_p$ were between 40 and 100 per cent for uniform and gamma renewal processes about 25 per cent.)

ϕ x 10^{-1}	n x 10^{-2}	Emp. Var. x 10^3	Rel. Err.	d	0	Bias x 10^3				Mean squared error x 10^6				$\hat{\text{Varp}}$ x 10^3	Rel. Err. %
						1	2	5	0	1	2	5			
2	1	6,1	19	—	,27	,32	,14	—	,67	1,2	1,4	4,0	15,8	11	76
2	2	3,8	23		,52	,50	,42		,77	,40	,77	,97	2,03	5,4	43
2	4	1,7	12		,023	,079	,044	—	,035	,018	,039	,059	,198	2,7	64
2	10	,58	14	—	,077	—	,068	—	,084	,007	,006	,010	,020	1,1	86
8	1	7,0	36		1,87	1,36	1,30		,68	5,3	6,1	7,3	11,8	5,6	20
8	2	3,4	28		1,03	,89	,96		,79	1,22	,94	1,24	1,41	2,8	19
8	4	1,5	24		,185	,061	,006	—	,140	,13	,13	,20	,28	1,4	9
8	10	,66	16		,148	,091	,065		,013	,025	,011	,010	,011	,56	15

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REFERENCES

- COX, D.R. (1965). On the estimation of the intensity function of a stationary point process. *J.R. Statist. Soc. B*, 27, 332-337.
- LINHART, H. (1964). *The statistical analysis of stationary point processes*. Paper read at the Conference, S.A. Statist. Assoc., Oct. 1964, Pretoria.
- LINHART, H. (1970). The gamma point process. *S.Afr. Statist. J.* 4, 1-17.
- LINHART, H. (1972). *Point processes, an assessment of theory and methods*. Proc. 4th Symp. Math. Statistics, Pretoria, 29-51.
- LINHART, H. (1975). The standard error of a histogram from a stationary time series. *S. Afr. Statist. J.* 9,
- LINHART, H. and ZUCCHINI, W.S. (1970). Experiments on estimation in gamma point processes with exponential covariance. *S. Afr. Statist. J.*, 6, 19-26.
- LINHART, H. and ZUCCHINI, W.S. (1972). Parametric models for point processes. *S. Afr. Statist. J.* 6, 19-26.
- ZUCCHINI, W.S. (1973): *Estimating the intensity function of stationary point processes*. M.Sc. Thesis, Univ. of Natal, Durban.

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